AI for angiographic sequence analysis: research gaps and mitigation strategies

115

included studies

of 5,064 screened

82.6%

use temporal information, mostly shallowly

12 gaps · 14 strategies

C1 Data infrastructure

Research gaps — *studies*

- **G1** Public benchmarks — 64
- **G2** Domain shift — 67
- **G5** Annotation scarcity — 49
- **G10** Privacy & multi-site learning — 8

Mitigation strategies

- **S2** Challenge benchmarks
- **S3** Self-supervised pretraining
- **S4** Federated learning
- **S5** Domain adaptation
- **S8** Label-efficient learning
- **S14** Generative augmentation

C2 Modeling & learning

Research gaps — *studies*

- **G3** Temporal modeling — 61
- **G4** 2D limits & 3D reconstruction — 46
- **G9** Rare pathology & imbalance — 41

Mitigation strategies

- **S3** Self-supervised pretraining
- **S6** Video Transformers & SSMs
- **S7** Physics-informed 3D models
- **S8** Label-efficient learning
- **S9** Uncertainty & OOD detection
- **S14** Generative augmentation

C3 Clinical deployment

Research gaps — *studies*

- **G6** Prospective validation — 43
- **G7** Explainability — 12
- **G8** Multi-modal integration — 15

Mitigation strategies

- **S8** Label-efficient learning
- **S9** Uncertainty & OOD detection
- **S10** Prospective multi-center RCTs
- **S11** Outcome-supervised training
- **S12** Longitudinal outcome cohorts
- **S13** Vision-language report models

C4 Outcome & reporting

Research gaps — *studies*

- **G11** Outcome prediction — 9
- **G12** LLMs & structured reporting — 2

Mitigation strategies

- **S10** Prospective multi-center RCTs
- **S11** Outcome-supervised training
- **S12** Longitudinal outcome cohorts
- **S13** Vision-language report models

S1 Large multi-institutional annotated datasets (all clusters)

Bars: included studies whose full text documents each gap. Shared strategies appear under every cluster they address.